Extended reading — Week 13: Multi-Agent RL and AlphaZero

November 8, 2026

Read with a question in mind

Start with the week's teaching guide and its small worked example. Then use these readings to connect the algorithm to its assumptions and evidence. The explanations and investigations below are course commentary, not quotations. Read the source before reusing a theorem or a reported result. For each paper, record the objective, observation model, claimed result, conditions and one condition your local experiment does not reproduce.

1. Silver et al. (2017): Mastering the Game of Go without Human Knowledge

Primary source or book record: https://doi.org/10.1038/nature24270.

What to learn. AlphaGo Zero learns a policy and value from self-play, using search-derived policy targets and game outcomes.

What the result requires. Self-play predates this work. The achievement does not mean no game structure or engineering knowledge enters the representation, rules and search. A small Tic-Tac-Toe project illustrates the loop without reproducing its scale.

Notebook investigation. Use `exercise_2_terminal_value`. Name the player whose perspective each stored value represents, then explain the sign flip when backing a leaf value to its parent.

2. Silver et al. (2018): A General Reinforcement Learning Algorithm that Masters Chess, Shogi, and Go through Self-Play

Primary source or book record: https://arxiv.org/abs/1712.01815.

What to learn. AlphaZero adapts the search-and-learning approach across three games. It differs from AlphaGo Zero in several training details, including symmetry augmentation.

What the result requires. Success on these games is empirical evidence, not a proof for every perfect-information zero-sum game. Chess is not solved, and domain-specific legal moves and state/action encodings remain necessary.

Notebook investigation. Compare root noise settings in the source: Dirichlet alpha is 0.3 for chess, 0.15 for shogi and 0.03 for Go. Explain why legal-action count affects the role of that concentration parameter.

3. Yu et al. (2022): The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games

Primary source or book record: https://arxiv.org/abs/2103.01955.

What to learn. Studies PPO-based methods on particle-world environments, SMAC, Google Research Football and Hanabi. Implementation and training choices substantially affect the empirical comparison.

What the result requires. This is not a theorem of dominance or a result on MA-MuJoCo. A central critic can use privileged training information; decentralised actors must retain their permitted observation inputs.

Notebook investigation. Use `exercise_6_gae` and `exercise_7_coord_game_return`. Check time, environment and agent axes, then distinguish a critic input from an execution-time actor input.

4. Rashid et al. (2018): QMIX

Primary source or book record: https://arxiv.org/abs/1803.11485.

What to learn. Constrains a mixing network to be monotone in per-agent utilities so decentralised greedy choices can maximise its joint value representation.

What the result requires. The consistency property restricts the representable joint values. It does not establish globally optimal learning in every cooperative game or make QMIX a permanent state-of-the-art baseline.

Notebook investigation. Write a two-agent payoff table whose preferred action for one agent reverses when the other changes action. Explain why a fixed local utility ordering cannot represent that dependence monotonically.

5. Lanctot et al. (2017): A Unified Game-Theoretic Approach to Multiagent Reinforcement Learning

Primary source or book record: https://arxiv.org/abs/1711.00832.

What to learn. PSRO maintains a population of policies and trains responses to a meta-strategy over opponents. It connects practical RL response oracles with game-theoretic population methods.

What the result requires. Approximate neural best responses do not inherit every convergence property of exact finite double-oracle algorithms. A restricted-game equilibrium may still be exploitable by a missing policy.

Notebook investigation. For rock-paper-scissors, compare responding only to the latest opponent with responding to a mixture. State whether the evaluation pool contains every possible deviation.

6. Vinyals et al. (2019): Grandmaster Level in StarCraft II

Primary source or book record: https://doi.org/10.1038/s41586-019-1724-z.

What to learn. Combines large-scale learning, human-game pretraining and a league with distinct training roles to diversify opponents.

What the result requires. League diversity is evidence of a useful training design, not a guarantee of robustness against the entire strategy space. An evaluation win rate depends on the opponent population and rules.

Notebook investigation. Extend the workbook evaluation table to several fixed opponents and record wins, draws and losses. A score against random play is not an exploitability estimate.

7. Schrittwieser et al. (2020): Mastering Atari, Go, Chess and Shogi by Planning with a Learned Model

Primary source or book record: https://arxiv.org/abs/1911.08265.

What to learn. MuZero learns representations and latent dynamics used to predict rewards, values and policies during planning.

What the result requires. The model need not reconstruct observations or identify the true physical dynamics. The results do not imply an accurate latent rollout under every policy or a fixed sample-efficiency gain for later extensions.

Notebook investigation. Draw one latent transition and label its immediate reward, next latent state, policy and value predictions. Contrast this with the exact Tic-Tac-Toe transition used in the local MCTS.

Synthesis: evidence you should be able to show

1. Reproduce one numerical calculation before running a long experiment.
2. Name the precise objective, sampling distribution and terminal convention.
3. Compare the local implementation with one source: list a shared mechanism and one simplification that prevents calling it a full reproduction.
4. Separate an algebraic identity, an assumption-dependent theorem and an empirical result. A plot alone does not prove a theorem.
5. Save the notebook with seed, parameters, axes, denominators and a short explanation of an unexpected result. Keep evaluation data separate from tuning.

Return to the teaching guide's notebook investigation for the local function names, expected checks and interpretation questions. Source code and notebooks are available together in the course repository; the web reader is the entry point and the downloaded notebook is the execution workspace.

Statistics reading bridge

A win rate depends on the role distribution. Read weighted means and paired comparisons before pooling game results. Explain the role mixture, independent match unit and opponent protocol; keep AlphaZero and multi-agent primary readings for search and self-play claims.

Checked textbook locations: Siegrist § § 4.1, 4.5, PDF pp.439, 478; Ross § 7.3, PDF p.309. https://stats.libretexts.org/@go/page/10160. These locations support the probability and estimation concepts; the algorithm-specific results remain attributed to the primary readings above. The course source catalog distinguishes the original textbook from its supplied instructor manual and records PDF-viewer pagination.